



RECOMMENDED PRACTICE
RP B401
CATHODIC PROTECTION DESIGN

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1. INTRODUCTION

Cathodic protection implies that corrosion protection is achieved by making the item to be protected the **cathode** in an electrochemical cell. Corrosion of the protected object is thereby eliminated or reduced to an acceptably low level.

The process of cathodic protection requires an electrically conductive medium (**electrolyte**) and a current source. The latter may be either a corroding metal; i.e. a **sacrificial anode**, or an external direct current (D.C.) source. Sacrificial anodes are sometimes called "galvanic anodes". Cathodic protection by means of a D.C. rectifier is referred to as **impressed current cathodic protection**. Cathodic protection using sacrificial ('galvanic') anodes is referred to as **sacrificial anode cathodic protection**.

Cathodic protection is applicable to all structural metals and alloys. It is efficient in preventing general corrosion as well as localized forms of corrosion on both ferrous and non-ferrous alloys. Galvanic corrosion due to combination of dissimilar metals can also be prevented.

Marine cathodic protection covers the corrosion protection of external surfaces exposed to seawater or marine sediments, and of internal surfaces in compartments flooded by seawater. Steel reinforcement in submerged concrete structures is also included.

2. SCOPE

This document provides *detailed requirements* for the *design* and *fabrication (installation)* of marine cathodic protection systems utilizing sacrificial anodes. Additionally, more *general requirements* to the specification of sacrificial anode **manufacturing** are given.

Guidelines for the **inspection and monitoring** of cathodic protection during operation are offered in an *advisory* form. This also applies to the *guidelines* for design, installation and operation of impressed current cathodic protection systems.

This recommended practice is applicable to subsea pipelines as well as to all types of fixed offshore structures, including permanently installed floating installations.

3. DEFINITIONS, SYMBOLS AND ABBREVIATIONS

3.1 PARTIES AND RESPONSIBILITIES

Operator party responsible for the operation of a cathodically protected object

Designer party responsible for the design of a cathodic protection system

Fabricator party responsible for the installation of a cathodic protection system

As this recommended practice covers *all* phases of cathodic protection, from conceptual engineering to operation, it is normally pertinent to refer to the applicable chapters for a certain application; e.g. chapters 6 and 7 for detailed engineering and Chapter 8 for anode manufacturing.

An Operator may decide to overrule specific requirements to detailed design and fabrication in this document, making them more or less conservative. Any such deviation shall be communicated to the Designer in writing, e.g. in a document defining the design premises for a specific project. Any exceptions to mandatory requirements which are initiated by the Designer or Fabricator shall be approved by the Operator in writing.

3.2 USE OF LANGUAGE

In chapters 6, 7, 8 and 9 the wording "shall" and "is to be" are *mandatory* requirements of this recommended practice. Similarly, "should" in the same chapters indicates a non-mandatory advice, while "may" refers to an option.

3.3 GENERAL TECHNICAL TERMS

References within brackets refer to definitions given of chapter/section/paragraph of this document.

anode	electrode at which electrical current enters the electrolyte	
bracelet anode		(5.8.1)
cathode	electrode at which electrical current leaves the electrolyte	
cathodic disbonding		(5.5.1)
cathodic protection		(1)
corrosion allowance		(5.2.3)
dielectric shield		(10.4.5)
electrode	electronic conductor through which an electrical current can enter or leave an electrolyte	
electrolyte	electrically conducting medium in which the current is transported by ions	
flush mounted anode		(5.8.1)
galvanic anode	synonymous to <i>sacrificial anode</i>	
impressed current cathodic protection		(1)
polarization	change in potential of an <i>electrode</i> caused by a current load	
sacrificial anode		(1)
stand-off anode		(5.8.1)

3.4 DESIGN PARAMETERS AND SYMBOLS

References within brackets refer to chapter/section/paragraph of this document.

anode current capacity, c_a (A·h)	(7.8.4)
anode current output, I_a (A)	(7.8.2)
anode length, L (m)	(6.7)
anode resistance, R_a (ohm)	(6.7)
anode surface area, A_a (m ²)	(6.7)
anode utilization factor, u	(6.9)
average (maintenance) design current density, i_c (average) (A/m ²)	(6.3.5)
cathode surface area, A_c (m ²)	(7.4.2)
closed circuit anode potential, E_a° (V rel Ag/AgCl/seawater)	(6.6.5)
coating category	(6.4.4)
coating breakdown factor, f_c	(6.4.1)
current demand, I_c (A)	(7.4.1)
current drain (A)	(7.5)
design life, t_r (yrs)	(6.2.1)
resistivity, ρ (ohm·m)	(6.8)
driving voltage, ΔE (V)	(7.8.2)
electrochemical efficiency, ϵ (A·h/kg)	(6.6.2)
final design current density i_c (final) (A/m ²)	(6.3.3)
initial design current density, i_c (initial) (A/m ²)	(6.3.2)
number of anodes, n	(7.8.4)
protection potential, E_c (V rel Ag/AgCl/seawater)	(5.4.4)
protective potential, E_c° (V rel Ag/AgCl/seawater)	(5.4.1)
net mass of individual anode, m (kg)	(7.8.4)
total anode current capacity, C_a (A·h)	(7.8.4)
total anode net mass, M (kg)	(7.7)
unit	(7.2)

3.5 ABBREVIATIONS

BS	British Standard
D.C.	Direct Current
DFT	Dry Film Thickness
DNV	Det Norske Veritas
HISC	Hydrogen Induced Stress Cracking
I.C.	Impressed Current
ISO	International Organization for Standardization
NACE	National Association of Corrosion Engineers (U.S.A)
NDE	Non Destructive Examination
ROV	Remotely Operated Vehicle
RP	Recommended Practice
SMYS	Specified Minimum Yield Strength

4. REFERENCES

- NACE Standard Recommended Practice RP0176-83:
"Corrosion Control of Steel Fixed Offshore Platforms Associated with Petroleum Production".
- NACE Standard Recommended Practice RP0387-87:
"Metallurgical and Inspection Requirements for Cast Sacrificial Anodes for Offshore Applications".
- NACE Standard Recommended Practice RP0492-92:
"Metallurgical and Inspection Requirements for Offshore Pipeline Bracelet Anodes".
- BS 7361 (1991):
"Cathodic Protection Part 1. Code of Practice for Land and Marine Applications".
- ISO 8501-1/SIS 055900-1989:
"Pictorial Surface Preparation Standards for Painting Steel Surfaces".
- U.S. Military Specification MIL-A-18001J (1983 + Amendment 1, 1987):
"Anodes, Corrosion Preventive, Zinc; Slab, Disc and Rod Shaped".
- "Design and Operational Guidance on Cathodic Protection of Offshore Structures, Subsea Installations and Pipelines". The Marine Technology Directorate Ltd., London 1990.

5. GENERAL DESIGN CONSIDERATIONS

5.1 GENERAL

The detailed design of offshore structures and pipelines will normally be preceded by a *conceptual design* activity including corrosion protection schemes. Typically, this will involve materials selection for major components, specification of coating utilization, choice of cathodic protection concept (e.g. sacrificial versus impressed current), selection of anode material and type, etc. This activity will also consider functional requirements with respect to reliability, maintenance and monitoring.

This chapter is intended as a *guideline* for Operators and to Designers in the conceptual phase of defining the premises for the detailed engineering of cathodic protection systems. Some of the advice given in this chapter may have consequences for other disciplines, e.g. structural, marine operations and weight control.

It is emphasized that nothing in this chapter shall be considered as a mandatory requirement related to chapters 6, 7, 8 and 9.

5.2 LIMITATIONS OF CATHODIC PROTECTION

5.2.1 For carbon and low-alloy steels, cathodic protection should be considered as a technique for orrosion control, rather than to provide full immunity. It follows that cathodic protection is not an alternative to corrosion resistant alloys for components with very high dimensional tolerances, e.g. sealing assemblies.

5.2.2 Cathodic protection may not be fully effective under conditions favouring severe bacterial corrosion in areas efficiently shielded from cathodic protection, e.g. in narrow crevices or under disbonded coatings.

5.2.3 For critical structural or pressure containing components with long design lives, it may be pertinent to consider some extra thickness ("**corrosion allowance**") for inclusion in the design to allow for some residual corrosion.

5.2.4 Special measures may be required to prevent corrosion damage prior to installation at sea. For impressed current cathodic protection systems this will also apply to the period from installation until the system has been energized.

5.3 ENVIRONMENTAL PARAMETERS AFFECTING CATHODIC PROTECTION

5.3.1 The major seawater parameters affecting cathodic protection are:

- dissolved oxygen content
- sea currents
- temperature
- marine growth
- salinity

These parameters are interrelated and vary with geographic location. In addition, they are often subject to major variations with depth and season.

5.3.2 The parameters listed above determine the cathodic current density demand to *achieve* and to *maintain* cathodic protection. It is, however, not feasible to give any exact relation between these environmental parameters and the cathodic current demand. To rationalize the design of cathodic protection systems, **design current densities** have been defined based on climatic zones and depth (6.3).

The ambient seawater temperature and salinity are decisive to the seawater resistivity (6.8) and hence to the anode resistance (6.7) which determines the anode current output (7.8).

5.4 PROTECTIVE POTENTIALS

5.4.1 A potential of **-0.80 V** relative to the silver/silver chloride/seawater (Ag/AgCl/seawater) reference electrode is generally accepted as the **protective potential (E_c)** for carbon steel and low-alloy steel in aerated seawater.

5.4.2 In **anaerobic** environments including typical marine sediments, a lower protective potential, **-0.90 V** rel. Ag/AgCl/seawater, has widely been considered necessary to achieve efficient corrosion protection.

5.4.3 The protective potentials above are applicable as *criteria* for monitoring that corrosion protection has been achieved (see Chapter 11).

5.4.4 It should be noted that the *optimum* protection potential (E_c), i.e. the actual potential resulting in the lowest current consumption, is typically in the range -0.90 to -1.00 V rel. Ag/AgCl/seawater. This potential will be achieved spontaneously for a correctly designed cathodic protection system based on sacrificial anodes.

5.5 DETRIMENTAL EFFECTS OF CATHODIC PROTECTION

5.5.1 Cathodic protection will be accompanied by the formation of hydroxyl ions and hydrogen at the surface of the protected object. These products may cause disbonding of coatings (**cathodic disbonding**) by various mechanisms including chemical dissolution and electrochemical reduction processes at the steel/coating interface, sometimes in combination with the build-up of hydrogen pressure at this interface.

Coating systems normally applied for marine applications (e.g. epoxy, vinyl and polyurethane based products) are usually resistant to cathodic disbonding down to a potential of -1.1 V rel. Ag/AgCl/seawater.

Cathodic protection by means of impressed current or magnesium anodes may lead to more negative potentials than -1.1 V rel. Ag/AgCl/seawater. Coatings to be applied where the potential may be more negative than -1.1 V shall have documented resistance to cathodic disbonding.

5.5.2 Cathodic protection will cause discharge of hydrogen atoms at the metal surface. Some of this hydrogen will become absorbed into the metal matrix. In combination with high tensile loads and/or high internal stresses, this hydrogen can cause **hydrogen induced stress cracking (HISC)** of susceptible materials.

5.5.3 Ferrous and non-ferrous alloys with Specified Minimum Yield Strength (SMYS) up to 550 MPa are considered fully compatible with cathodic protection systems utilizing aluminium or zinc based sacrificial anodes. Any welding or other fabrication process affecting mechanical properties shall be carried out according to a qualified procedure which limits the maximum hardness to HV10 350 (Vickers hardness).

5.5.4 Quenched and tempered low-alloy steels, cold-worked austenitic stainless steels, precipitation hardened ferrous and non-ferrous alloys with SMYS up to 720 MPa that are **not** intended for welding, are considered compatible with sacrificial anode cathodic protection systems. It is recommended that the actual yield strength and hardness is limited to maximum 850 MPa and HV10 350 respectively (see also paragraphs 5.5.5 and 5.5.6 below).

5.5.5 Certain nickel-base alloys (e.g. UNS N05500) have suffered severe HISC at SMYS lower than 720 MPa. The susceptibility has been attributed to work hardening operations prior to a precipitation hardening heat treatment.

5.5.6 It is advisable to carry out qualification testing whenever materials with SMYS in excess of 550 MPa are to be used for specially critical applications, or in excessively severe conditions for which practical experience is lacking. Such applications may involve plastic strain, particularly when combined with cyclic loads, or protective potentials more negative than -1.1 V rel. Ag/AgCl/seawater.

5.5.7 Qualification testing should attempt to simulate the most severe loading conditions and environmental factors anticipated in practice. Crack Tip Opening Displacement (CTOD) tests and Constant Extension Rate Testing (CERT) carried out in an appropriate environment and at a controlled protection potential are relevant for this purpose. General testing procedures for CTOD and CERT testing are described in ISO 7539-7 and BS:5762 respectively.

5.5.8 A consequence of cathodic protection is that a **calcareous layer** will form on bare metal surfaces. The thickness of such layers is typically of the order of a tenth of a millimeter but thicker deposits may occur.

In general, these calcareous layers reduce current demand and have no harmful effects. However, there is some risk that they may obstruct mating of subsea electrical or hydraulic connectors with small tolerances. This may be prevented by applying an insulating layer of a thin film coating (e.g. baked epoxy resin).

5.5.9 Cathodic protection will eliminate the **anti-fouling properties** of copper-base alloys in seawater.

5.6 IMPRESSED CURRENT VERSUS SACRIFICIAL ANODE SYSTEMS

5.6.1 Application of impressed current (I.C.) cathodic protection requires a D.C. supply.

5.6.2 The major advantage of I.C. systems compared to sacrificial anode systems is that the number and size of anodes can be reduced, therewith reducing weight and drag forces.

5.6.3 An efficient spread of current from a small number of anodes in an I.C. system normally requires a coating system and preferably also a simple geometry of the object to be protected. Hulls of floating exploration/production platforms is an important application for I.C. systems.

5.6.4 I.C. systems have found application for *retrofitting* (or upgrading) of cathodic protection systems based on sacrificial anodes. They can also be applied *in combination with* sacrificial anodes.

5.6.5 I.C. anodes and lead wires are highly vulnerable to mechanical damage during installation and operation. Anode to lead wire electrical connections is another critical link.

5.6.6 The life of I.C. anodes is not unlimited. Application of I.C. systems for fixed offshore installations will require that procedures for subsea maintenance and repair are developed and qualified.

5.6.7 Reliability and costs for installation, maintenance and monitoring should be taken into account when choosing between I.C. and sacrificial anode cathodic protection systems.

5.6.8 I.C. systems can readily give protection potentials more negative than -1.1 V rel. Ag/AgCl/seawater which may cause damage to materials and coatings (see Section 5.5). This may have implications for the selection of structural materials and coating systems. Stringent on-line control of the protection potential is considered a prerequisite for the application of I.C. cathodic protection.

5.6.9 I.C. systems are not suitable for closed tanks and other non-vented closed spaces, because potentials more negative than -1.1 V (Ag/AgCl/seawater) may lead to hydrogen evolution and explosion hazards. Chlorine is another hazardous gas produced at I.C. anodes.

5.6.10 Regulations may require that I.C. system shall be shut off during diver operations and certain explosive operations (e.g. well perforations).

5.7 SACRIFICIAL ANODE MATERIALS

5.7.1 Sacrificial anodes for offshore applications are based on either aluminium or zinc. The generic type of anode material to be used is normally specified by the Operator. Al-base anode materials are mostly preferred due to their higher electrochemical efficiency. Mg-base anodes have been used in a few cases in combination with Al-base anodes to achieve rapid polarization during the initial phase.

5.7.2 Many anode materials are proprietary alloys. The Operator may require that the anode manufacturer shall document the service performance (6.6) by offshore operational experience or by long-term testing in natural seawater. A recommended testing procedure for long-term testing of anode materials is given in Appendix B.

5.7.3 Recommended maximum contents of impurity elements are given in Table 5.7.

Table 5.7 Recommended maximum concentrations of impurity elements in Zn- and Al-based anode materials.

Impurity Element	Maximum content (% weight)	
	Zn-base	Al-base
Fe	0.005	0.10
Cu	0.005	0.006
Pb	0.006	-
Si	0.12	0.15

5.7.4 Certain sacrificial anode alloys contain additions of mercury or cadmium. Statutory regulations limiting the use of sacrificial anode alloys may apply.

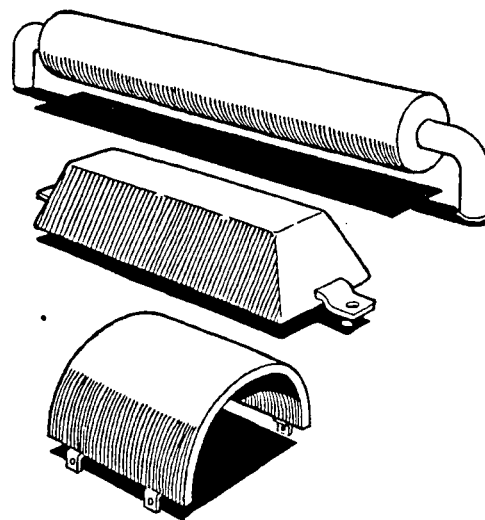
5.8 ANODE GEOMETRY

5.8.1 The major types of sacrificial anode design (see Figure 5.8) for offshore applications are:

- slender stand-off
- elongated, flush mounted
- bracelet

The type of anode design to be applied is normally specified by the Operator, and should take into account various factors, including:

- anode utilization and current output
- restrictions for installation and subsea intervention procedures
- weight and drag forces exerted by sea currents
- costs for manufacturing and installation
- ability to entangle flexibles, e.g. diver hoses or ROV umbilicals



The slender stand-off anode type has the highest current output and utilization factor. However, such anodes are obstructive and cause high drag forces in surface waters.

Fig. 5.8 Some major types of anode design: slender stand-off (top), elongated flush mounted (middle) and half-shell bracelet (bottom)

5.8.2 Requirements to sacrificial anode cores and fastening devices are given in Section 7.9.

5.9 USE OF COATINGS IN COMBINATION WITH CATHODIC PROTECTION

5.9.1 The application of non-metallic coatings reduces the current demand for cathodic protection and hence the required weight of sacrificial anodes. In particular, for weight-sensitive structures with long design lives, the combination of coatings and cathodic protection may give the most cost-effective corrosion protection.

5.9.2 Coatings are also beneficial in achieving a rapid polarization and promoting a more effective current distribution for objects with complex geometries. For impressed current cathodic protection systems, the utilization of coatings is normally a prerequisite to achieve adequate spread of the protective current.

5.9.3 Application of a coating system will reduce the risk of bacterial corrosion of C-steel and low-alloy steel in areas shielded from cathodic protection current, e.g. in narrow crevices.

5.9.4 Coating systems applied in combination with cathodic protection are to be selected based on documented practical experience. When such experience is lacking, a comprehensive qualification testing program is recommended.

5.9.5 The application of coatings may not be suitable for parts of submerged structures requiring frequent inspection for fatigue cracks; e.g. critical welded nodes of jacket structures.

5.9.6 Thermally sprayed aluminium coatings, usually in combination with a thin organic topcoat ('sealer'), have been applied for certain applications involving cathodic protection. Testing has shown that such coatings, with and without a sealer, markedly reduce the current demand for cathodic protection.

5.9.7 Zinc coatings and zinc-rich primers without a topcoat are unsuitable for the purpose of decreasing the current demand for cathodic protection.

5.9.8 Certain pipeline coatings (primarily those having high contents of carbon black) may have a significant electric conductivity allowing discharge of cathodic current at the coating/seawater interface. It is recommended that the volume resistivity shall be minimum $1.0 \cdot 10^4$ ohm·cm.

5.10 ELECTRICAL CONTINUITY AND CURRENT DRAIN

5.10.1 Provisions for electrical insulation of interfacing cathodic protection systems are in general unnecessary and are sometimes detrimental. One exception is interfaces between impressed current and sacrificial anode systems at landfalls of subsea pipelines where electrical insulation is usually justified.

Electrical insulation is sometimes installed between interfacing systems operated by different parties.

5.10.2 Special provisions may be required to ensure **electrical continuity** within a cathodic protection system (see Section 7.11).

5.10.3 The possibility of **current drain** from a cathodic protection system to neighbouring structures without cathodic protection or with deficient cathodic protection systems shall be considered during the design (see also Section 7.5).

5.11 DOCUMENTATION OF DESIGN PREMISES

Prior to the start of detailed design of a cathodic protection system, the **design premises** should be defined in a dedicated document. This document is to be approved by the Operator and shall include all relevant design parameters as specified by the Operator or selected by the Designer.

6. DESIGN PARAMETERS

6.1 GENERAL

This chapter specifies parameters to be applied in the design of cathodic protection systems based on sacrificial anodes. Some of the parameters are also applicable to impressed current cathodic protection systems (see Chapter 10).

With the exception of the design life (6.2), it is often assumed that the design parameters shall be identified/selected by the Designer of a cathodic protection system.

The design data given in this document are consistently selected from a *conservative* approach. Adherence to these data is therefore likely to provide an actual life which exceeds the design life of the system in most cases. Operators' own experience or other special considerations may justify the application of less, or in certain cases more conservative design data (see Section 3.1).

6.2 DESIGN LIFE

6.2.1 The **design life** (t_d) of a cathodic protection system is to be specified by the Operator. When specifying the design life of an object, the Operator should also consider the likelihood of the design life of the object being extended.

6.2.2 Maintenance and repair of sacrificial anode cathodic protection systems for fixed offshore structures and subsea pipelines is generally very costly and often impractical. It is therefore normal practice to apply the same anode design life as for the object to be protected.

6.2.3 In certain circumstances planned retrofitting of sacrificial anodes may be an economically viable alternative to installation of very large anodes initially. This alternative should be planned such that necessary provisions for retrofitting are made during initial design and fabrication.

6.3 DESIGN CURRENT DENSITIES

6.3.1 **Current density** refers to the cathodic protection current per unit surface area. The **initial** and **final design current densities**, i_c (initial) and i_c (final), give a measure of the anticipated cathodic current density demands to *achieve* cathodic protection of *bare* metal surfaces within a reasonably short period of time. They are used to calculate the initial and final current demands (7.4), which determine the number and sizing of anodes.

The effect of any coating is taken into account by application of a coating breakdown factor (see sections 6.4 and 6.5).

6.3.2 The **initial** design current density refers to the cathodic current density that is required to effect polarization of an initially exposed bare metal surface, typically with some atmospheric rusting and/or mill scale. The initial current density is necessarily higher than the average final current density because the calcareous scale developed during this initial phase reduces the subsequent current demands. A proper initial design current density enables rapid formation of protective calcareous scale and hence efficient polarization.

6.3.3 The **final** design current density refers to metal surfaces with established marine growth and calcareous layers. It takes into account the additional current density required to re-polarize the structure if such layers are partly and periodically damaged, e.g. by severe storms, impacts or cleaning operations. It follows that the final design current density is lower than the initial.

6.3.4 Both the initial and final design current densities refer to the cathodic current density at a potential of -0.80 V rel. Ag/AgCl/seawater *during the dynamic polarization* from a more positive (i.e. "non-protective") potential towards the protective potential range.

6.3.5 The **average** (or *maintenance*) design current density is a measure of the anticipated cathodic current density, once the cathodic protection system has attained its *steady-state* protection potential. This will imply a lower driving voltage and the average design current density is therefore lower than both the initial and final design values.

As the initial polarization period proceeding 'steady-state' conditions is normally quite short compared to the design life, the time-weighted current density becomes very close to the steady state current density.

The average (maintenance) design current density is used to calculate the minimum mass of anode material necessary to maintain cathodic protection throughout the design life (7.7).

6.3.6 The design current densities are dependent on factors that vary with geographical location and operational depth (see Section 5.3). Recommendations for design current densities are given in Table 6.3.1 and Table 6.3.2. These design current densities have been selected in a conservative manner to include harsh weather conditions, including waves and sea currents, but not, however, any erosional effects from dispersed silt or ice. They further take into account that the seawater is saturated with oxygen (i.e. at 0.21 bara) and the effects of marine fouling.

The data for 'Temperate' regions are applicable to the North Sea and the Norwegian Sea up to 70°N .

6.3.7 For bare steel surfaces **buried in sediments** a design current density (initial/final and average) of 0.020 A/m² is recommended irrespective of geographical location and depth.

6.3.8 For pipes and other components heated by an internal fluid, the design current densities (initial/final and average) as specified in Tables 6.3.1/6.3.2 and Paragraph 6.3.7 shall be increased by 0.001 A/m² for each $^{\circ}\text{C}$ that the metal/environment interface is assumed to exceed 25°C . For single wall conduits this temperature shall be assumed to be equal to the temperature of the internal fluid.

Table 6.3.1 Initial and final design current densities for various *climatic regions* and *depths*. The definitions of climatic regions in this table are based on a range of yearly *average* surface water temperatures as given in the same table.

Depth (m)	Design Current Densities (initial/final) in A/m ²			
	Tropical ($> 20^{\circ}\text{C}$)	Sub-Tropical ($12-20^{\circ}\text{C}$)	Temperate ($7-12^{\circ}\text{C}$)	Arctic ($< 7^{\circ}\text{C}$)
0 - 30	0.150	0.170	0.200	0.250
	0.090	0.110	0.130	0.170
> 30	0.130	0.150	0.180	0.220
	0.080	0.090	0.110	0.130

6.3.9 The current demands for freely flooded compartments and for closed compartments with free access to air will be limited by the dissolved oxygen content. Application of data for depths > 30 m in Table 6.3.1. are recommended.

Table 6.3.2 Average (maintenance) design current densities as a function of depth and climatic region (see Table 6.3.3)

Depth (m)	Design Current Densities (average) in A/m ²			
	Tropical (> 20°C)	Sub-Tropical (12-20°C)	Temperate (7-12°C)	Arctic (< 7°C)
0 - 30	0.070	0.080	0.100	0.120 ¹⁾
> 30	0.060	0.070	0.080	0.100

¹⁾ Effects of any ice scouring are not included.

6.3.10 It is becoming increasingly recognized that cathodic protection of **concrete reinforcing steel** is necessary to ensure long-term integrity of submerged concrete structures. However, even in cases where corrosion protection of reinforcement is not considered necessary, any anodes dedicated to the corrosion protection of components mounted on/embedded in the external concrete surfaces must also be designed to account for current *drain* from the cathodic protection system to the reinforcement.

6.3.11 The cathodic current density of steel embedded in concrete is mainly controlled by the reduction of oxygen. This is transported by capillary action of pore water driven by evaporation in the atmospheric zone and in internal dry compartments. The cathodic current density will therefore be dependent on depth and climatic conditions.

6.3.12 Table 6.3.3 gives design current densities applicable as both initial/final and average design values. It is to be noted that they refer to the total surface area of concrete reinforcing *steel*, **not** to the surface area of concrete.

Table 6.3.3 Design current densities (initial/final and average) for concrete reinforcing steel

Elevation (m)	Design Current Densities in A/m ²			
	Tropical (> 20°C)	Sub-tropical (12-20°C)	Temperate 7-12°C)	Arctic (< 7°C)
+5 to -10	0.0030	0.0025	0.0015	0.0010
< -10	0.0020	0.0015	0.0010	0.0008

6.3.13 For light weight aggregate concrete or other concrete grades with equivalent pore structure, the design current densities in Table 6.3.3 can be reduced by 40%.

6.3.14 When the actual reinforcing steel surface area (m²) to reinforced concrete volume (m³) ratio **B** exceeds 5, an adjustment factor 5/B may be applied to the design current densities in Table 6.3.3.

6.3.15 For reinforcing steel coated with fusion bonded epoxy or equivalent (min 200 µm DFT), see Paragraph 6.4.10.

6.3.16 For surfaces with a **thermally sprayed aluminium coating** (200 μm nominal thickness) a value of 0.010 A/m² is to be applied as average design current density. This value is applicable to all conditions covered by Table 6.3.2. As the free corrosion potential of thermal sprayed aluminium is more negative than -0.80 V rel. Ag/AgCl/seawater, initial/final design current densities as defined in paragraphs 6.3.1 to 6.3.4 shall not apply.

6.3.17 For internally heated components with an external thermally sprayed aluminium coating, the design current density shall be increased by 0.0002 A/m² for each °C that the metal/seawater interface is assumed to exceed 25°C (compare Paragraph 6.3.8).

6.4 COATING BREAKDOWN FACTOR AND PAINT COATINGS

6.4.1 The **coating breakdown factor (f_c)** describes the anticipated reduction in cathodic current density due to the application of an electrically insulating coating. $f_c = 0$ means that the coating is 100% electrically insulating, thus decreasing the cathodic current density to zero. $f_c = 1$ implies that the coating has no protective properties, i.e. the cathodic current density of an initially coated surface is the same as for a bare steel surface.

6.4.2 It should be noted that the coating breakdown factor should not be confused with coating degradation as apparent by visual examination. A coating showing extensive blistering may still retain high electrically insulating properties. Conversely, an apparently perfect coated surface may allow a significant passage of current.

6.4.3 The coating breakdown factor is a function of coating properties, operational parameters and time. The coating properties are dependent on coating materials, surface preparation and the coating application.

The coating breakdown factor (f_c) can be described as

$$f_c = k_1 + k_2 \cdot t$$

where t is the coating life time in years, k_1 and k_2 are constants that are dependent on the coating properties.

6.4.4 For practical reasons, four **paint coating categories** have been defined to facilitate correlation of the coating breakdown factor to coating properties:

- | | | |
|---------------------|---|---|
| Category I | : | One layer of primer coat, about 50 μm nominal DFT (Dry Film Thickness) |
| Category II | : | One layer of primer coat, plus <i>minimum</i> one layer of intermediate top coat, 150 to 250 μm nominal DFT |
| Category III | : | One layer of primer coat, plus <i>minimum</i> two layers of intermediate/top coats, minimum 300 μm nominal DFT |
| Category IV | : | One layer of primer coat, plus <i>minimum</i> three layers of intermediate top coats, minimum 450 μm nominal DFT |

The constants k_1 and k_2 used for calculating the coating break-down factors are given in Table 6.4.1.

In Figs. 6.4.1 to 6.4.4 coating break-down factors are presented as a function of time in a graphical form.

6.4.5 It is assumed that coating damage is mainly dependent on marine growth (incl. cleaning operations to remove such growth) and the erosional effects of waves and currents, all of which are related to depth. For simplicity, the effect of seawater temperature has been excluded.

6.4.6 It has further been assumed that the applied coating system is suitable for submerged marine applications, that all coating work has been carried out in accordance with the coating manufacturer's recommendations, and that surface preparation has included blast cleaning to minimum SA 2.5 in accordance with ISO 8501-1.

For coatings applied *without* blast cleaning (incl. machined, ground and cold-rolled surfaces), a coating break-down factor $f_c = 1$ shall be applied in calculations of average and final current demands, while the initial demand shall be calculated as for Category I.

6.4.7 Documented long-term experience of subsea paint coating systems is scarce, hence the coating breakdown factors given in Paragraph 6.4.8 below are estimated on a conservative basis. It is appreciated that the definitions of Coating Categories according to Paragraph 6.4.4 do not account for either the generic type of coating or more complex paint coating systems. Operator's experience of a specific paint coating system may justify the use of less conservative coating break-down factors than specified in this document.

6.4.8 For cathodic protection design purposes the **average** and **final coating breakdown factors** are to be calculated by introducing the design life (t_f):

$$f_c (\text{average}) = k_1 + k_2 \cdot \frac{t_f}{2}$$

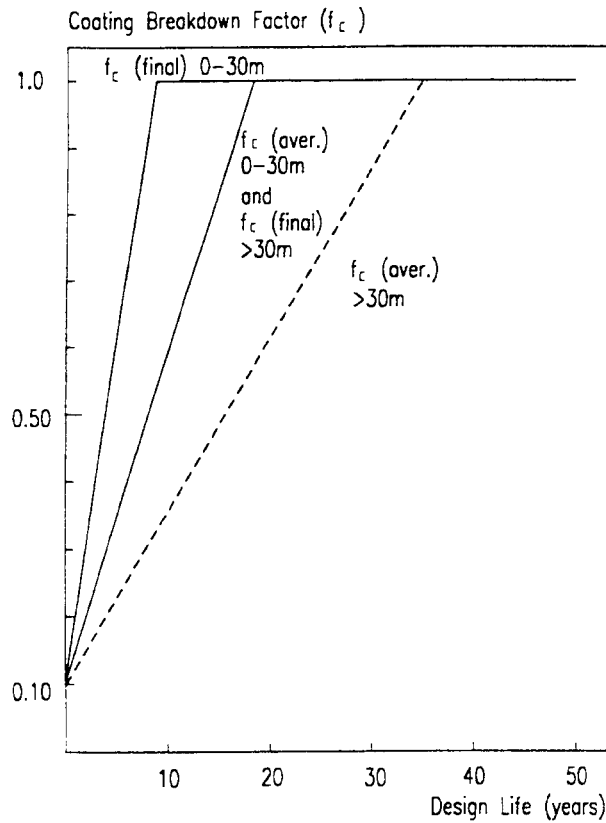
$$f_c (\text{final}) = k_1 + k_2 \cdot t_f$$

If the calculated value exceeds 1, $f_c = 1$ shall be applied in the design.

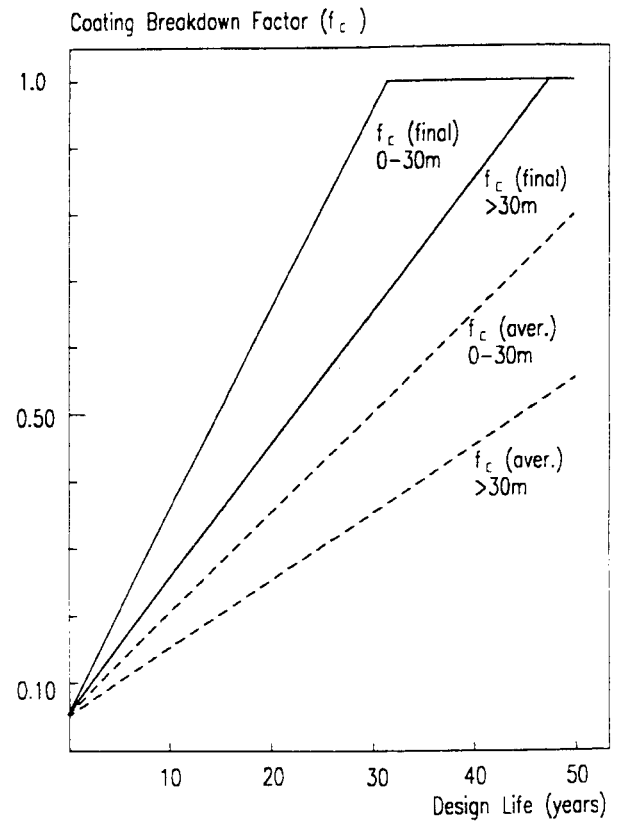
It is appreciated that the linear equations above are intended for the purpose of design only, and are not intended to reflect the true mode of coating breakdown.

When the design life of the cathodic protection system exceeds that of the coating system, f_c (average) may be calculated as:

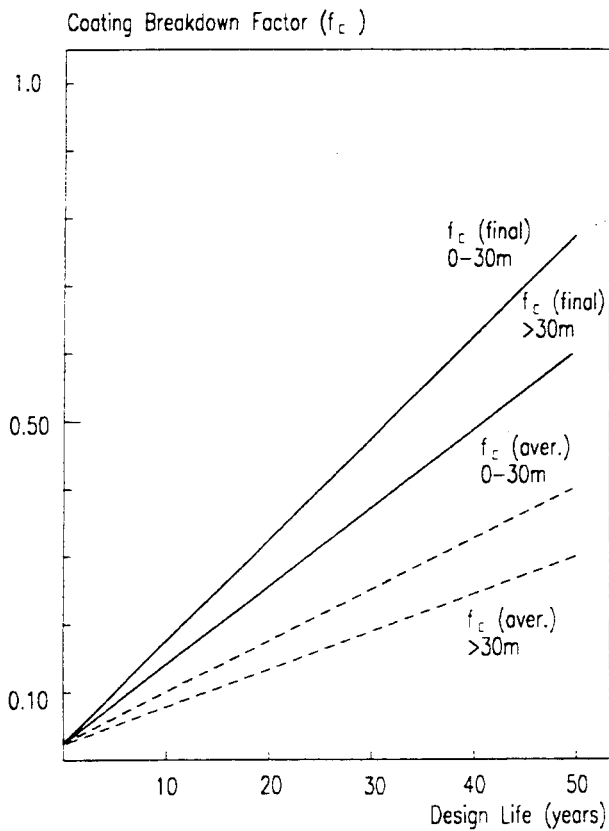
$$f_c (av.) = 1 - \frac{(1-k_1)^2}{2k_2 t_f}$$



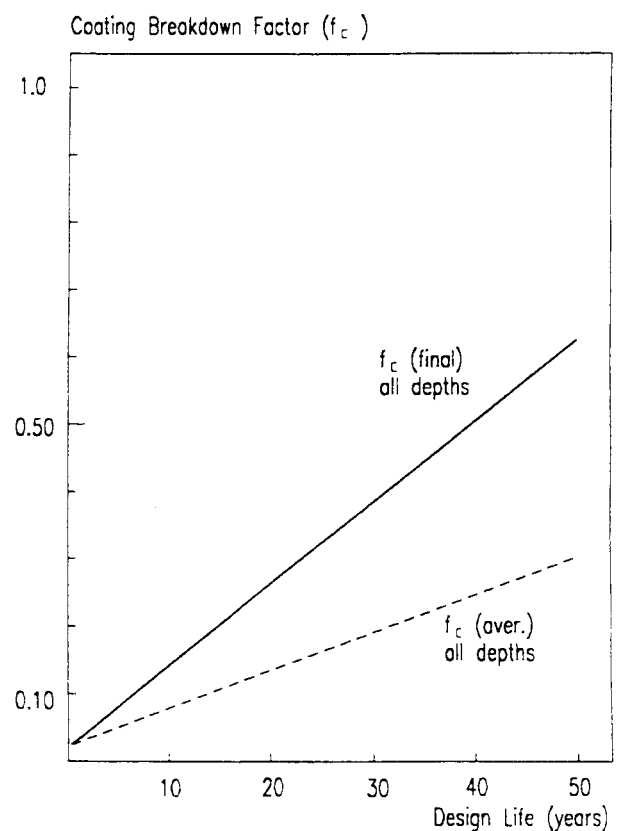
Coating Category I



Coating Category II



Coating Category III



Coating Category IV

Fig. 6.4.1 to 6.4.4. Graphical representation of coating breakdown factor as a function of time.

Table 6.4.1 Constants (k_1 and k_2) for calculation of paint coating breakdown factors. Coating Categories are defined in Paragraph 6.4.4.

Depth (m)	Coating Category			
	I ($k_1=0.10$) k_2	II ($k_1=0.05$) k_2	III ($k_1=0.02$) k_2	IV ($k_1=0.02$) k_2
0 - 30	0.10	0.03	0.015	0.012
> 30	0.05	0.02	0.012	0.012

6.4.9 Data for depths > 30 m may be applied to flooded compartments and to closed compartments with free access to air.

6.4.10 Data for Coating Category III, depth > 30 m, are applicable to calculations of cathodic current density demands for concrete reinforcing steel coated with fusion bonded epoxy or equivalent (min. 200 μ m DFT).

6.4.11 The coating breakdown factors as defined above do not include any allowance for damage to paint coatings during fabrication or installation. If such damage is anticipated to be significant, the affected surface area is to be estimated and included in design calculations (7.3/7.4) as bare metal surface.

6.5 PIPELINE COATINGS

6.5.1 The coating systems commonly applied for subsea pipelines and associated risers are highly efficient at reducing the current demand for cathodic protection. It is emphasized, however, that the economic consequences of a marginal cathodic protection design leading to unanticipated retrofitting of anodes and/or repairs of corrosion damage, are exceedingly high compared to the initial installation costs of a pipeline cathodic protection system. It is also common practice to upgrade the design life of subsea pipeline systems. Furthermore, the possibilities of monitoring pipeline cathodic protection are limited. Consideration of these factors justifies a *conservative* design of pipeline cathodic protection systems.

6.5.2 The coating breakdown factors (f_c) as defined in 6.4 and specified below for pipeline coatings shall be applied for design calculations unless otherwise specified or approved by the Operator. The recommended coating breakdown factors are applicable to pipeline coating systems based on:

- asphalt + concrete weight coating
- fusion bonded epoxy + adhesive + polyethylene or polypropylene
- polychloroprene rubber
- equivalent coating systems based on an inner layer dedicated to corrosion protection and one or more outer layer(s) for mechanical protection, possibly in combination with gravity or thermal insulation functions.

It is further assumed that pipe coatings and field joint coating systems have been selected to be suitable to the maximum design temperature of the pipeline, and that they are applied according to specifications reflecting good industrial practice.

6.5.3 The following pipeline coating breakdown factors relating to coating systems according to Paragraph 6.5.2 are applicable to both buried and non-buried pipelines:

$$f_c (\text{average}) = 0.05 + 0.002 (t_r - 30)$$

$$f_c (\text{final}) = 0.07 + 0.004 (t_r - 20)$$

A graphical presentation is given in Fig. 6.5.1.

6.5.4 For pipelines coated with fusion bonded epoxy (min 300 μm DFT) *without* any overwrap, the coating breakdown factors for Category III in Section 6.4 are applicable, modified to $k_1 = 0.05$.

6.5.5 There is a lack of long-term experience for many pipeline coating systems, hence the coating breakdown factors given above have been specified to ensure a conservative design, and do not purport to reflect the actual coating degradation. Operator's experience of specific pipeline coating systems may justify the application of less conservative coating breakdown factors.

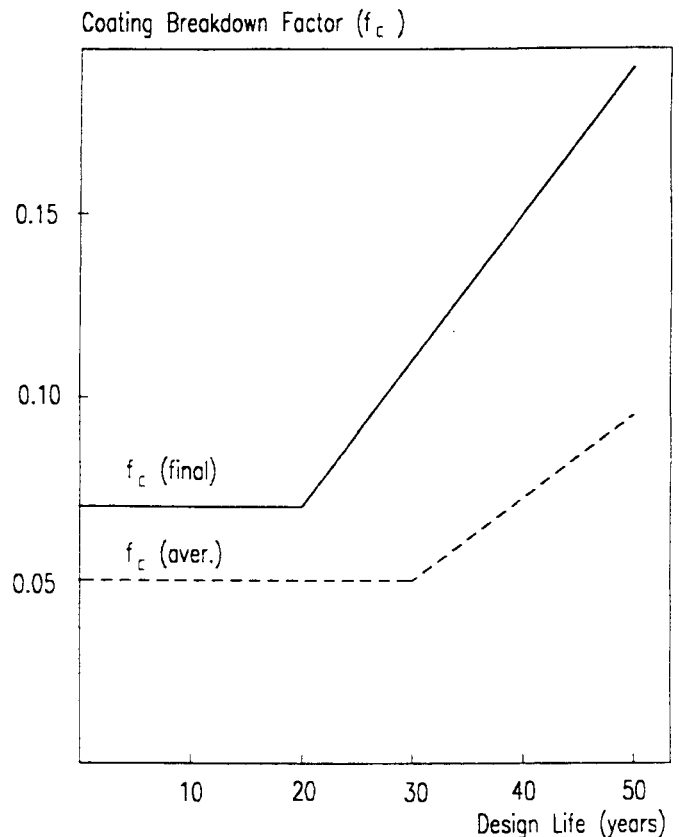


Fig. 6.5.1 Graphical representation of pipeline coating breakdown factor as a function of time.

6.6 SACRIFICIAL ANODE MATERIAL PERFORMANCE

6.6.1 The performance of a sacrificial anode material is dependent on its actual chemical composition (see Section 5.7). Normally the anode manufacturer has not been nominated at the design stage and no detailed anode performance data are available. This requires that more general and conservative design data are used.

6.6.2 Table 6.6.1 gives the **electrochemical efficiency** (ϵ) of anode materials *to be applied in design calculations* for anode weight requirement. It is assumed that the maximum contents of impurity elements of the anode material will not exceed the limits given in Table 5.7.

Table 6.6.1 *Design* Electrochemical Efficiency values for Al and Zn based sacrificial anode materials

Anode Material Type	Electrochemical Efficiency (Ah/kg)
Al-base	2000 ¹⁾
Zn-base	700 ²⁾

1) Anode temperature: max 25°C

2) Anode temperature: max 50°C

6.6.3 Use of higher values than those given in Table 6.6.1 shall be subject to approval by the Operator (see sections 3.1 and 5.7). However, it is recommended that electrochemical efficiencies used in design calculations based on the data in Table 6.3.2 shall never exceed 2500 Ah/kg and 750 Ah/kg for Al-base and Zn-base materials respectively.

6.6.4 For buried pipelines carrying hot fluids, heating of anodes by the internal medium will decrease the electrochemical efficiency of Al-base anodes. Reliable performance data for heated anodes are scarce. Fig. 6.6.1 may be used as a *guideline* as to the reduction of electrochemical efficiency with temperature.

For zinc anodes that are to be buried in sediments, the same electrochemical efficiency as for seawater may be applied for internal fluid temperatures up to 50°C. Higher temperatures may lead to intergranular corrosion and loss of protective properties.

Special zinc base sacrificial anode materials for application exceeding 50°C have been developed. The electrochemical efficiency to be employed for such alloys shall be subject to Operator's approval.

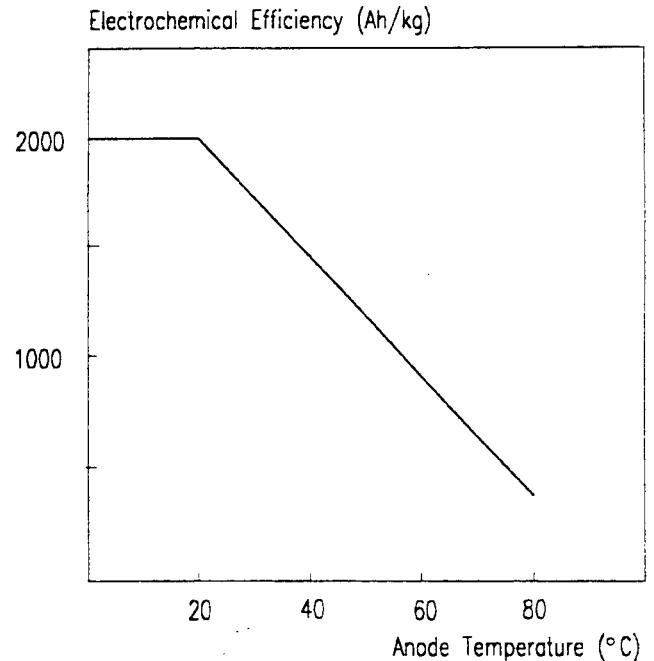


Fig. 6.6.1 Electrochemical efficiency (ϵ) of Al-base anode material as a function of anode temperature (T), i.e. $\epsilon = 2000 - 27(T-20)$.

6.6.5 The closed circuit anode potential (E_a) used in design calculations for determination of anode current output (7.8) shall not exceed the values in Table 6.6.2.

Table 6.6.2 *Design Closed Circuit Anode Potentials for Al and Zn based sacrificial anode materials.*

Anode Material Type	Environment	Closed Circuit Anode Potential (V rel. Ag/AgCl/seawater)
Al-base	Seawater	- 1.05
	Sediments	- 0.95
Zn-base	Seawater	- 1.00
	Sediments	- 0.95

6.7 ANODE RESISTANCE FORMULAS

6.7.1 Table 6.7.1 gives recommended anode resistance formulas for various anode shapes of sacrificial anodes. Unless otherwise specified by the Operator, the formulas in Table 6.7.1 are to be applied to each relevant anode design in the design calculations (7.8).

The seawater resistivity to be used is defined in Section 6.8.

Table 6.7.1 Anode Resistance (R_a) Formulas

Anode Type	Resistance Formula
Long slender stand-off ¹⁾ $L \geq 4r$	$R_a = \frac{\rho}{2 \cdot \pi \cdot L} \left(\ln \frac{4L}{r} - 1 \right)$
Short slender stand-off ¹⁾ $L < 4r$	$R_a = \frac{\rho}{2 \cdot \pi \cdot L} \left[\ln \left\{ \frac{2L}{r} \left(1 + \sqrt{1 + \left(\frac{r}{2L} \right)^2} \right) \right\} + \frac{r}{2L} - \sqrt{1 + \left(\frac{r}{2L} \right)^2} \right]$
Long flush-mounted $L \geq 4 \times \text{width and thickness}$	$R_a = \frac{\rho}{2 \cdot S}$
Short flush-mounted, bracelet and other flush-mounted shapes	$R_a = \frac{0.315 \cdot \rho}{\sqrt{A}}$

Abbreviations:

ρ (ohm·m)	= environmental resistivity
L (m)	= length of anode
r (m)	= radius ²⁾ of anode
S (m)	= arithmetic mean of anode length and width
A (m ²)	= exposed anode surface area

- 1) The equation is valid for anodes with minimum distance 0.30 m from object. For anode-to-object distance less than 0.30 m but minimum 0.15 m the same equation may be applied if a correction factor 1.3 is applied.
- 2) For non-cylindrical anodes: $r = C/2\pi$ where $C(m)$ = cross sectional periphery

6.7.2 To calculate the initial anode resistance, R_a (initial), the initial anode dimensions are to be inserted into the relevant anode resistance formula of Table 6.7.1.

The final anode resistance, R_a (final), i.e. the resistance when the anode has been consumed to its utilization factor (6.9), shall be assessed as recommended in Paragraphs 6.7.3 to 6.7.7 below.

6.7.3 All anode types:

When the anode has been consumed to its utilization factor u , the remaining anode mass m is given by:

$$m \text{ (final)} = m \text{ (initial)} \cdot (1-u)$$

The initial and final volume of the anode can be calculated from the anode mass, density and volume of insert materials. When details of anode inserts are not available, their volume may be either deleted or estimated to provide a conservative approach.

6.7.4 Slender stand-off anodes:

Assume a length reduction corresponding to 10% of the net anode mass/volume reduction when the anode has been consumed to its utilization factor (see Section 6.8), i.e.

$$L \text{ (final)} = L \text{ (initial)} - 0.10 \cdot u \cdot L \text{ (initial)}$$

where L (final/initial) refer to the anode length. Assume the final shape is a cylinder. Calculate the final length and radius according to Paragraph 6.7.3 above.

6.7.5 Long Flush-Mounted Anodes:

Assume that the final shape is a semi-cylinder. Calculate the final length and radius (= width) as in Paragraph 6.7.4 above.

6.7.6 Half-Shell Bracelet Anodes:

For pipeline half-shell bracelet anodes with ends mounted flush with concrete weight coating and gaps filled with mastic, calculate the final exposed surface area from the initial dimensions and net mass, and the utilization factor in Table 6.9.1.

6.7.7 Short Flush-Mounted Anodes and "Other Shapes" with anode resistance calculated from the formula

$$R_a = \frac{0.315 \cdot \rho}{\sqrt{A}}$$

Assume that the final exposed surface area is equivalent to the initial surface area of the anode facing the surface to be protected.

6.8 RESISTIVITY

6.8.1 The resistivity ρ ($\text{ohm} \cdot \text{m}$) of seawater is a function of its salinity and temperature. In the open sea the salinity does not vary significantly and temperature is the main factor. The relation between resistivity and temperature at a salinity of 30 to 40‰ (parts per thousand) is given in Fig. 6.8.

6.8.2 At shore areas, particularly in the vicinity of river outlets and in enclosed bays, the salinity will vary significantly. It is recommended that the design of cathodic protection systems in such locations is based on resistivity measurements.

6.8.3 Compared to seawater, the resistivity of marine sediments is higher by a factor ranging from 2 for very soft clays to approximately 5 for sand. Unless actual sediment data for the location are available, the highest factor is to be assumed.

6.8.4 In temperate regions (average surface water temperature 7 to 12°C) resistivities of 0.30 and 1.5 $\text{ohm} \cdot \text{m}$ are recommended as conservative estimates for the calculation of anode resistances in seawater and in marine sediments respectively. Lower values are to be documented by actual measurements, taking into account any seasonal variations in temperature.

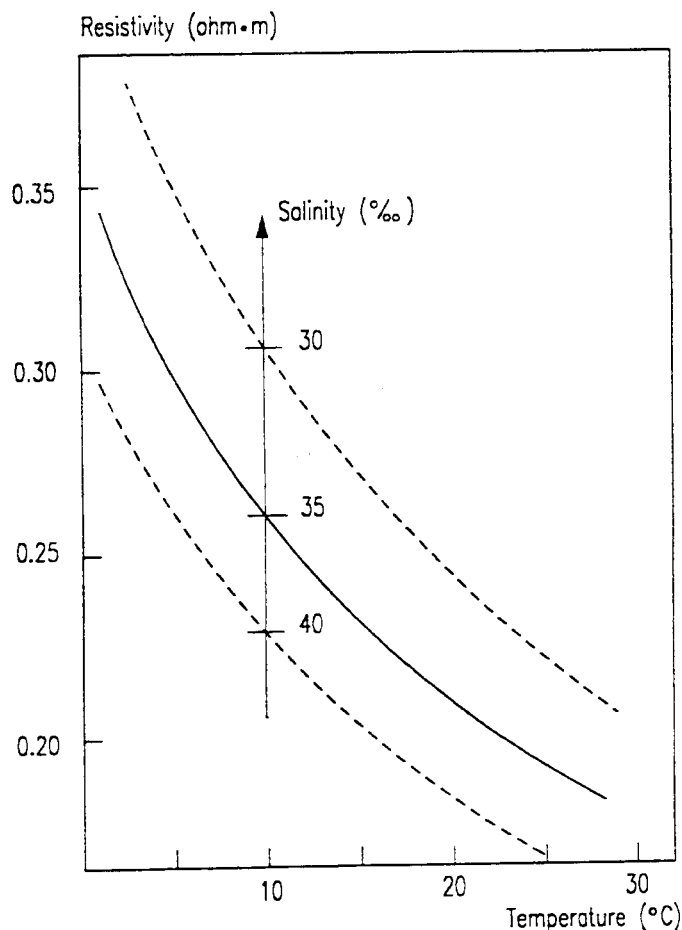


Fig. 6.8 Seawater resistivity as a function of temperature for salinity 30 to 40‰.

6.9 ANODE UTILIZATION FACTOR

6.9.1 The **anode utilization factor (u)** expresses the fraction of anode material that may be utilized for design purposes. When the anode is consumed beyond the utilization factor, the anode performance becomes unpredictable due to loss of support of anode material and/or rapid increase of anode resistance.

6.9.2 The utilization factor of an anode is dependent on the detailed anode design, in particular with respect to dimensions and location of anode cores. The utilization factors in Table 6.9.1 are conservative estimates independent of detailed anode design and shall be applied for design calculations, unless otherwise has been specified or approved by the Operator.

Table 6.9.1 Design Anode Utilization Factors

Anode Type	Anode Utilization Factor
Long ¹⁾ slender stand-off	0.90
Long ¹⁾ flush-mounted	0.85
Short ²⁾ flush-mounted	0.80
Bracelet, half-shell type	0.80
Bracelet, segmented type	0.75

1) Anode length $\geq 4 \times$ thickness

2) Anode length $< 4 \times$ thickness

7. DESIGN TASKS

7.1 GENERAL

The sequence of design tasks adopted in this chapter may not always be the most convenient. The sequence may thus be changed to suit individual cases, but *all* the design tasks shall be completed for the detailed design of a cathodic protection system based on sacrificial anodes.

It is essential that the engineering calculations are adequately detailed and documented to allow checking and third party verification.

7.2 SUB-DIVISION OF OBJECT

7.2.1 In the design of cathodic protection systems for large and complex objects, it is always convenient, and often necessary to sub-divide the object into units to be protected.

7.2.2 The division into units may be based on environmental or operational parameters, e.g. depth zones for a jacket structure or fluid temperature for a pipeline. It may also be determined by physical interfaces of the protection object, e.g. retrievable units within a subsea production system.

7.3 SURFACE AREA CALCULATIONS

7.3.1 For each unit (7.2) surface areas to receive cathodic protection shall be calculated separately for areas where the environmental conditions or the application of coatings imply different current requirements.

7.3.2 It is practical to apply some simplifications when calculating surface areas for objects with complex geometries. However, it shall be ensured that the overall result of any such simplifications is reasonably conservative.

7.3.3 Surface area calculations shall be documented in the design report. References shall be made to drawings, including revision numbers.

7.3.4 The Designer of a cathodic protection system shall make sure that *all* components to be connected to the system are included in the surface area calculations. This may include various types of outfitting to be installed by other contractors.

7.3.5 When details about "outfittings" (or degree of coatings) are not available at design of cathodic protection, it may be pertinent to include some extra surface area. The actual surface areas to be protected shall be checked at a later design stage.

7.3.6 It is sometimes necessary to account for various items temporarily installed and for temporary current drain during installation and work-over phases. (7.5)

7.4 CURRENT DEMAND CALCULATIONS

7.4.1 The **current demand** (I_c) to *achieve* polarization during the initial *and* final life of the cathodic protection system, and the average current demand to *maintain* cathodic protection throughout the design life shall be calculated separately.

7.4.2 The individual areas (A_c) of each unit to be cathodically protected (7.2) are to be multiplied with the relevant design current density (i_c), and the coating breakdown factor (f_c), if applicable:

$$I_c = A_c \cdot f_c \cdot i_c$$

where I_c is the current demand for a specific surface area. i_c is to be selected from Section 6.3. f_c refers to Table 6.4.1 and paragraphs 6.5.3/6.5.4 for paint coatings and pipeline coatings respectively.

7.4.3 For items with major surface areas of bare (uncoated) metal, the current demands required for initial polarization, I_c (initial), and for re-polarization at the end of the design life, I_c (final), are to be calculated, together with the average current demand I_c (average) required to maintain cathodic protection throughout the design period. For pipelines and other items with high quality coatings, the initial current demand can be deleted in the design calculations.

7.5 CURRENT DRAIN CALCULATIONS

7.5.1 More complex offshore structures often include temporary or permanent components which are not considered to require cathodic protection but will *drain* current from the system. Such items may be components used during installation only, or secondary structural components which can readily tolerate some corrosive wear. Also metallic materials with intrinsic resistance to corrosion in seawater will still drain current from the cathodic protection system.

7.5.2 All items which exert a significant current drain on the cathodic protection system are to be included in the surface area and current demand calculations as specified in sections 7.2 to 7.4.

7.5.3 For components freely exposed to seawater, the design current densities in Tables 6.3.1 and 6.3.2 are to be applied. For current drain to concrete reinforcement, see Paragraph 6.3.10.

7.5.4 For parts of steel skirts and piles to be buried in sediments, a design current density (initial/final and average) of 0.020 A/m^2 is recommended as a conservative approach. Current drain to open pile internals shall cover 10 x diameter.

7.5.5 Unless otherwise specified by the Operator, a current drain of 5A shall be included for each well when well casings are a part of the object to be protected.

7.5.6 For interfacing cathodic protection systems (e.g. template/pipeline/platform) designed according to this document, no current drain between individual systems needs to be included in the design calculations.

7.6 SELECTION OF ANODE TYPE AND DIMENSIONS

7.6.1 The type of anode (5.8) to be utilized is largely dependent on fabrication, installation and operational parameters. Consequently the anode type is normally specified by the Operator. The anode type is determining for which anode resistance formulas (6.7) and anode utilization factors (6.9) are used in further calculations.

7.6.2 For pipeline bracelet anodes that are mounted flush with the coating, the thickness of the coating layer will be decisive to the anode dimensions.

7.7 ANODE MASS CALCULATIONS

The **total net anode mass** M (kg) required to maintain cathodic protection throughout the design life t (yrs) is to be calculated from I_c (**average**) for each unit of the object (including any current drain).

$$M = \frac{I_c(\text{average}) \cdot t \cdot 8760}{u \cdot \epsilon}$$

where ϵ (A·h/kg) is the electrochemical efficiency of the anode material (6.6), u is the utilization factor (6.9) and 8760 refers to hours per year.

7.8 CALCULATION OF NUMBER OF ANODES

7.8.1 For the anode type selected (7.6) the number of anodes, anode dimensions and anode net mass shall be selected to meet the requirements for initial/final current output (A) *and* the current capacity (A·h) which relate to the protection current demand of the protection object.

7.8.2 The **anode current output** (I_a) is calculated from Ohm's law:

$$I_a = \frac{E_c^\circ - E_a^\circ}{R_a}$$

where E_a° (V) is the *design* closed circuit potential of the anode (6.6.5), R_a (ohm) is the anode resistance (6.7) which is assumed to be equivalent to the total circuit resistance. E_c° (V) is the *design* protective potential which is chosen to **-0.80 V** (Ag/AgCl/seawater).

$\Delta E^\circ = E_c^\circ - E_a^\circ$ is referred to as the *design driving voltage*.

Note: As the initial and final design current densities in Section 6.3 refer to a protection potential of **-0.80 V** (Ag/AgCl/seawater) during the initial polarization and any transient depolarization/re-polarization, $E_c^\circ = \text{b-0.80 V}$ shall always be used for design calculations. This applies also to anaerobic environments when the *protective* potential (5.4) has been chosen to **-0.90 V** (Ag/AgCl/seawater). The latter potential is then applicable as a *criterion* for monitoring that adequate protection has been achieved, but does not affect the design calculations.

7.8.3 The anode current output is to be calculated for the initial *and* final life of the cathodic protection system. In the latter case anodes shall be assumed to be consumed to their utilization factor. Guidelines for calculation of final anode resistances are given in Section 6.7.

7.8.4 The anode current capacity (c_a) is given by

$$c_a = m \cdot \epsilon \cdot u$$

where m is the net mass per anode. The total current capacity (C_a) thus becomes $n \cdot c_a$ (A·h) where n is the number of anodes.

7.8.5 For pipelines without burial, rock dumping or trenching specified, the anode current output (I_a) and current capacity (C_a) may be calculated as if all of the anodes were surrounded by seawater only.

7.8.6 Anode dimensions and net weight are to be selected to match all requirements for current output (initial/final) and current capacity for a specific number of anodes. This is an *iterative* process and a simple computer spreadsheet may be helpful.

7.8.7 Calculations shall be carried out to demonstrate that the following requirements are met:

$$C_a = n \cdot c_a \geq I_c (\text{average}) \cdot t_r \cdot 8760$$

and

$$n \cdot I_a (\text{initial/final}) \geq I_c (\text{initial/final})$$

where I_c is the cathodic current demand of the object or unit thereof (7.4), including any current drain (7.5).

7.9 DETAILED ANODE DESIGN

7.9.1 Final selection of anode design should involve contacts with potential vendors of sacrificial anodes to ensure that the specified design will be commercially available.

7.9.2 Anode cores, supports and fastening devices shall be designed to provide the required utilization factor, to ensure electrical continuity and to support the anode during all phases of fabrication, installation, and operation of the object to be protected. Doubler and/or gusset plates may be required (see Section 9.1).

7.9.3 To avoid slippage of pipeline anodes during laying, anodes to be mounted on to top of the pipeline coating are preferably to be designed for direct welding of anode core to doubler plates on the pipeline.

7.9.4 Special considerations are required for anodes on structural members subject to pile driving operations during installation. Strength analyses may be required to ensure adequate resistance to fatigue loads.

7.9.5 Flush mounted anodes shall be paint coated on surfaces facing the object to be protected. 100 μm of coal tar epoxy is adequate.

7.9.6 An estimate of the remaining anode dimensions when consumed to the utilization factor, as applied in the design (see Section 6.9), shall be performed to check that significant exposure of the anode core is unlikely.

7.9.7 Detailed anode drawings including fastening devices, shall be prepared for each anode type/size. Connection cables shall be detailed where applicable. Net and gross anode weights including tolerances are to be specified on drawings.

7.10 DISTRIBUTION OF ANODES

7.10.1 The calculated number of anodes are to be distributed to provide a uniform current distribution, taking into account the current demand of individual members due to different surface areas, any coatings used, etc.

7.10.2 Special considerations are required when anodes cannot be attached directly to certain members to obtain cathodic protection, e.g. conductors, tether pipes, various outfitings.

7.10.3 If practical, anodes dedicated to the protection of surfaces buried in sediments shall be located freely exposed to seawater.

7.10.4 Anodes shall preferably be located with sufficient spacing between each other to avoid interaction effects that reduce the useful current output.

7.10.5 The distance between successive pipeline anodes should not normally exceed 150 m.

7.10.6 The distribution of anodes shall take into account restrictions imposed by fabrication, installation and operation. Early liaison with other engineering disciplines, as well as with fabrication and installation contractors is advised.

7.10.7 Simple numerical techniques are sufficient to provide uniform current distribution. In special cases mathematical modelling may be helpful to verify optimum current distribution or possible interference of adjacent anodes.

7.11 ANODE MOUNTING AND PROVISIONS FOR ELECTRICAL CONTINUITY

7.11.1 Anodes are to be securely attached to the object, taking account of all relevant forces exerted during fabrication, installation and operation (see Section 7.9).

7.11.2 Direct welding of anode mounting devices to individual members of the object to be protected is the preferred method to obtain both structural integrity and electrical continuity.

7.11.3 With pressurised components or items subject to high external loads, welded anode supports or electrical connections shall be placed at least 150 mm away from other welds, and the need of any doubler plates shall be considered.

7.11.4 If anodes are to be clamped or bolted to the object, a stranded cable shall be employed to ensure the necessary electrical continuity between the anode insert and the object. The cable shall be selected to provide adequate strength and conductivity.

7.11.5 Pipeline bracelet anodes shall have minimum one stranded cable for each half-shell.

7.11.6 Attachment of anode connection cables may involve "thermite" welding, brazing, friction or explosion welding. The risk of copper penetration shall be considered if copper base materials are used for fusion.

Requirements to the connection, e.g. conductor configuration, qualification, inspection and testing, etc., shall be specified by the Designer.

7.11.7 When welding or brazing of connection cables is not feasible, mechanical connections using serrated washers, screws with spike tips, etc. may be used. The requirements to electrical resistance across such connections and the verification thereof, shall be specified. It is recommended that the product of the connection resistance multiplied by the maximum anode current output shall not exceed 10% of the design driving voltage (7.8.2).

7.11.8 For cathodic protection of multi-component assemblies without welded connections, electrical continuity may be assumed across any metal seals and non-coated threaded connections. For other components that shall have cathodic protection but lack dedicated anodes, the electrical continuity to one or more remote anodes shall be ensured using stranded cables as above.

7.11.9 The requirements for electrical resistance across bolted cable connections (7.11.8) shall be specified. It is recommended that the product of the connection resistance multiplied by the maximum current demand (initial or final) of the object shall not exceed 10% of the design driving voltage (7.8.2).

7.12 ENGINEERING DOCUMENTATION

7.12.1 The detailed engineering documentation shall contain the following items:

- *Design Premises*
incl. reference to relevant project specifications, codes, standards, etc.
- *Surface Area Calculations*
incl. reference to relevant drawings, incl. revision number.
- *Current Demand Calculations*
incl. initial/final and average
- *Current Drain Calculations*
if applicable
- *Anode Mass Calculations*
- *Anode Resistance Calculations*
- *Anode Number Calculations*
incl. anode current output and anode capacity for initial and final life of system.
- *Anode Detailed Design Drawings*
incl. fastening devices (and connector cables if applicable).
- *Anode Distribution Drawings*
- *Requirements to Anode Mounting and Provisions for Electrical Continuity*

7.12.2 The documentation shall be sufficiently well organized and detailed to allow third party verification.

8. MANUFACTURING OF SACRIFICIAL ANODES

8.1 GENERAL

Sacrificial anodes shall be manufactured according to a procedure which as a minimum includes:

- *chemical analysis of anode material*
The compositional ranges for alloying elements and the maximum allowable contents of all relevant impurity elements shall be specified. Sampling frequency is to be defined.
- *requirements to anode core materials and core preparation*
Inserts to be welded to the object shall be made of steel with adequate weldability. The carbon equivalent (CE) shall not exceed 0.45 when the following formula is applied:

$$CE = C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$$

- *Weight and dimensional tolerances*
- *Visual examination*
Defect tolerances shall be defined
- *Non-destructive and/or destructive examination*
(if applicable)
- *Marking and documentation*
- *Electrochemical testing for quality control*

8.2 QUALIFICATION OF ANODE MANUFACTURING

The Operator may require that anode manufacturing shall be carried out according to a qualified procedure.

8.3 STANDARDS

Requirements to manufacturing of anodes are given in NACE RP 0492-92 and RP 0387-87 for pipeline and other offshore anodes respectively.

8.4 QUALITY CONTROL

A recommended procedure for electrochemical testing of anode material is given in Appendix A.

9. INSTALLATION OF SACRIFICIAL ANODES

9.1 GENERAL

9.1.1 All installation of anodes (and connector cables if applicable) shall be carried out according to specifications and drawings *approved for construction*.

9.1.2 Any subsequent changes of anode location or anode fastening devices (incl. connector cables) shall be approved by the Operator.

9.1.3 Prior to installation, the Fabricator shall inspect all anodes for any damage that may have occurred during transportation or handling. Anodes which do not conform to the anode manufacturing specification shall be rejected.

9.1.4 All welding/brazing of anode fastening devices and connector cables shall be carried out by qualified personnel according to a qualified procedure using the applicable fabrication construction codes. A procedure for visual inspection and NDE of welded/brazed connections shall be established by the Fabricator.

9.1.5 Verification of electrical continuity across any bolted cable connections shall be carried out according to a written procedure.

9.1.6 Installed anodes are to be protected during any subsequent coating work. All spillage of coating on anodes shall be removed. For coated objects, anode fastening devices should be coated to the same standard as for the object

9.2 PIPELINE ANODES

9.2.1 Pipeline anodes are normally of the half-shell bracelet type. The bracelets are clamped or welded to the pipe joints after application of the corrosion coating. Stranded connector cables are to be used for clamped half-shell anodes.

9.2.2 For anodes mounted on pipe joints that are concrete weight coated, measures shall be taken to avoid electrical contact between concrete reinforcement and the anodes.

The gaps between the anode half-shells shall preferably be filled with asphalt mastic or a similar compound. Any spillage of filling component on the external surface shall be removed.

9.3 DOCUMENTATION

9.3.1 The final anode distribution shall be indicated on *as-built* drawings.

9.3.2 Verification of electrical continuity across any bolted electrical connections shall be documented.

10 IMPRESSED CURRENT CATHODIC PROTECTION SYSTEMS

10.1 GENERAL

10.1.1 This chapter offers *guidance* for the design and installation of I.C. cathodic protection systems.

10.1.2 Some general design considerations applicable to I.C. systems are given in Chapter 5.

10.1.3 The design, installation and commissioning tasks of I.C. cathodic protection systems are normally carried out by the same contractor, here referred to as the Designer.

10.2 CURRENT SOURCES AND CONTROL

10.2.1 I.C. systems normally utilize a transformer-rectifier as the current source. Statutory requirements may apply for its design, installation and operation.

10.2.2 The transformer-rectifier shall have sufficient capacity to match the calculated current demand of the object to be protected (see Section 10.4). The needs for installation of spare capacity shall be duly considered. It is advisable to carry out a failure mode effect analysis.

10.2.3 The control of the transformer-rectifier current output may be manual or automatic. In the former case, either current or voltage control may be used while automatic control shall be based on potential readings from fixed reference electrodes. Alarm functions indicating excessive voltage/current load on anodes and/or too negative, or too high protection potentials should be included.

10.2.4 It is preferable that the current output from individual anodes can be varied independently of each other.

10.3 IMPRESSED CURRENT ANODE MATERIALS

10.3.1 The most commonly used I.C. anode material is platinum on a substrate of titanium, niobium or tantalum, typically applied by cladding, tack welding or electroplating, .

10.3.2 Anodes made of lead-silver alloy, graphite, magnetite, silicon iron and certain other materials are applicable as semi-inert I.C. anode materials but these have not been widely used for fixed offshore structures.

10.3.3 The choice of anode material and substrate will determine the maximum operating voltage and anodic current density that can be utilized.

10.3.4 I.C. anode materials will suffer some wear, the magnitude of which is dependent on the anodic current density and the applied voltage.

The Designer should be capable of documenting the long-term anode performance at maximum operating current/voltage in order to assess the probable life of anodes.

10.4 SYSTEM DESIGN

10.4.1 The design of an I.C. cathodic protection system should include some **redundancy** with respect to the number of anodes. This is to account for both a more uneven current distribution and a higher risk of anode malfunction compared to sacrificial anode systems.

It is recommended that the total maximum current output from all the anodes should be *minimum* a factor 1.5 higher than the total current demand based on i_c (final) as given in Table 6.3.1.

10.4.2 The current demand for the object to be protected shall be calculated as described in sections 7.2 to 7.5.

10.4.3 The optimum distribution of I.C. anodes is decisive to an efficient system. It is recommended that computerized mathematical modelling of current and potential distributions should be performed.

10.4.4 I.C. anodes shall be designed to withstand all forces that can reasonably be expected during installation and operation. Anodes shall also be designed to be replaced either from the surface, or subsea utilizing a diver or ROV.

10.4.5 Anodes to be mounted flush with the object to be protected, or at a minimum distance of less than 1.5 meters, shall have a relatively thick non-conductive coating or sheet ('dielectric shield') applied to the adjacent surface of the object. This is to prevent excessively negative potentials (i.e. more negative than -1.1 V rel. Ag/AgCl/seawater) and accompanying risks of HISC and coating damage (see Section 5.5).

Typically, shields of pre-fabricated glassfibre reinforced sheets of epoxy resin are used and may be fastened to the object with an epoxy resin.

10.4.6 Anode cables should have a steel armour and are to be adequately protected by routing within a dedicated conduit (or internally within the structure if applicable). Provisions for repairs and replacement of cables shall be designed for.

10.4.7 Restrictions for routing of cables in hazardous areas may apply (see Paragraph 10.2.1).

10.4.8 Due consideration shall be given to the design and materials selection for any subsea connectors and to cable terminations at anodes. Platinum or gold coated titanium, niobium and tantalum are adequate for electrically conducting interfaces.

10.4.9 For subsea pipeline I.C. systems, insulating joints are commonly applied to avoid interaction with sacrificial anode systems at platforms or at landfalls. Insulating joints shall be located above water at a location suitable for inspection and maintenance.

10.5 MONITORING EQUIPMENT

10.5.1 A sufficient number of fixed reference electrodes shall be installed to verify adequate cathodic protection of the entire object. This is most essential for any critical areas where inspection with divers or ROV is prevented.

10.5.2 Reference electrodes shall be designed for mechanical integrity during installation and operation. Provisions should be made for in-situ calibration and subsea replacement of reference electrodes.

10.5.3 Cables to reference electrodes shall be mechanically protected as for anode cables (see Paragraph 10.4.6) but shall not be placed in a common conduit. All cables to reference electrodes shall be screened to promote electrical shielding.

10.6 MANUFACTURING AND INSTALLATION CONSIDERATIONS

10.6.1 Stringent quality control during manufacturing of I.C. anodes, subsea electrical connectors and dielectric shields is essential. The same applies to mounting of anodes and dielectric shields.

10.6.2 Several vendors may be involved in the manufacturing and installation operations. The Designer should establish a Quality Surveillance Plan specific to each contract.

10.6.3 Installation of anodes, reference electrodes and electric shields on the object shall be carried out according to drawings approved for construction. All changes shall be subject to the Operator's approval.

10.6.4 Any welding or brazing of equipment to the structure shall be carried out according to qualified procedures.

10.7 DOCUMENTATION

10.7.1 As a minimum requirement the **detailed design documentation** shall contain the following items:

- Transformer-Rectifier documentation (capacity, control and alarm facilities, circuit drawings, etc.).
- Detailed drawings of anodes, reference electrodes, subsea connectors, cable terminations, dielectric shields and other equipment to be installed subsea.
- Procedures for fastening anodes, reference electrodes and dielectric shields (if applicable).
- Documentation of anode performance.
- Calculation of current demands.
- Drawings showing location of all anode and reference electrodes.
- Detailed drawings of cables (and electrically insulating joints if applicable).
- Drawings of cable protection conduits (and wall penetrations if applicable).
- Documentation of potential monitoring system.
- Quality Surveillance Plan.

The engineering documentation shall be sufficiently well organized and detailed to allow third party verification.

10.7.2 The **as-built documentation** shall contain all relevant engineering documentation updated to "as-built" status, and a detailed manual for operation and maintenance of the system.

11 INSPECTION AND MONITORING OF CATHODIC PROTECTION

11.1 GENERAL

11.1.1 Inspection and monitoring of cathodic protection is carried out to confirm that criteria for adequate corrosion protection are fulfilled, and to detect any general deficiencies of the cathodic protection system. A further objective may be to collect data for optimization of future cathodic protection designs.

11.1.2 This chapter is relevant to cathodic protection systems based on sacrificial anodes. A special feature of such systems is that they are self-regulating within certain limits. A marginal design with respect to number and/or sizing of anodes may lead to a high driving voltage, and increased anode current output. A consequence of this will be accelerated anode consumption and therewith reduced life of the system. Even though the life time is reduced the protection criteria may still be fulfilled for this period.

It is essential that any deficiencies of a cathodic protection system can be detected to assess the needs for both urgent and long-term measures for upgrading of the system.

11.2 DATA COLLECTION

11.2.1 **Periodic inspection** of cathodic protection is normally based on potential measurements (11.3) utilizing a standard reference electrode carried by a diver or a remotely operated vehicle (ROV), in combination with visual observations (11.4).

11.2.2 **On-line monitoring** of cathodic protection uses fixed instrumented reference electrodes. Additionally, provisions may be made for recording current output from some representative anodes.

11.3 POTENTIAL MEASUREMENTS

11.3.1 Probes for potential measurements are commonly based on an Ag/AgCl/seawater reference electrode, but other reference electrodes may apply.

11.3.2 Potential readings utilizing diver or ROV controlled probes are to be carried out according to a detailed procedure.

11.3.3 Potential readings should be preceded by on-site calibration of the reference electrode.

11.3.4 During measurements the reference electrode is to be positioned as close to the object as practical. This is most essential for measurements of anode potential, which may involve large voltage drops in the seawater adjacent to the anode.

11.3.5 Fixed reference electrodes may have a limited life and procedures for online calibration and maintenance should be established as required. Cables shall be electrically shielded and protected by pipe conduits or equivalent.

11.3.6 For carbon and low-alloy steels, any measurement of protection potentials more positive than -0.80V rel Ag/AgCl/seawater are indicative of underprotection and consequent potential risk of corrosion damage. Readings in the range -0.80 to -0.90V are indicative of marginal capacity of the cathodic protection system.

11.3.7 Anode potentials should preferably be more negative than the design closed circuit anode potential. More positive recordings are indicative of either excessive current load or passivation behaviour.

11.3.8 Combined measurements of anode potentials (E_a) and protection potentials (E_c) can be utilized to assess the anode current output (I_a) if the anode resistance (R_a) is estimated based on actual anode dimensions; i.e.

$$I_a = \frac{E_a - E_c}{R_a}$$

Based on the calculated current output, the remaining life of the anode can be estimated.

11.4 VISUAL INSPECTION

11.4.1 Direct visual inspection by divers or by cameras carried by ROV may disclose deficiencies of a cathodic protection system such as damaged or missing anodes, excessive anode consumption, etc.

11.4.2 Apparent corrosion damage or rust discolouration is another sign of underprotection. An abnormally low anode consumption may indicate passivation.

11.5 OTHER INSPECTION/MONITORING TECHNIQUES

11.5.1 There are several techniques for on-line measurements of sacrificial anode current output. Such data are applicable to calculations of anode remaining life. They are of special value for research and development activities aiming to optimize future designs.

11.5.2 Electric field measurements close to anodes and protected objects may be used for semi-quantitative calculations of anode current output and cathodic current density respectively.

11.5.3 Wall thickness measurements, e.g. utilization of intelligent pigs for pipelines, may disclose deficiencies of cathodic protection systems not otherwise accessible for either visual inspection or measurements of protective potential.

11.6 DESIGN AND FABRICATION

11.6.1 Subsea equipment for on-line monitoring of cathodic protection must be designed to provide long-term mechanical integrity and to be amenable to maintenance as required. Special consideration shall be given to protection of electrical cables.

11.6.2 All equipment shall be manufactured and installed according to drawings approved for construction.

11.7 EXTENT OF INSPECTION/MONITORING

11.7.1 The extent of inspection/monitoring of cathodic protection for a certain protected object must be assessed with due account of the various factors involved, incl. functional requirements, design, accessibility for other forms of inspection, etc. Governmental regulations may also apply.

11.7.2 Results from previous inspections are decisive for planning future inspection programs. Typically efforts shall be concentrated on any areas where marginal or insufficient cathodic protection has been identified during a previous survey and in areas where damage to anodes and/or coating has been reported.

11.8 DOCUMENTATION

11.8.1 All relevant documentation relating to the operation of on-line monitoring systems are to be collected in a dedicated manual.

11.8.2 Periodic inspection reports shall contain sufficient details of procedures and locations of measurements to optimize repeatability of recordings.

11.8.3 Historic data from periodic inspections and on-line monitoring are to be organized such that long-term trends of recorded parameters can be elucidated.

APPENDIX A

RECOMMENDED PRACTICE FOR ACCELERATED LABORATORY TESTING OF SACRIFICIAL ANODE MATERIALS WITH THE OBJECTIVE OF QUALITY CONTROL

A.1 GENERAL

A.1.1 This laboratory test procedure is intended for **quality control** of Al-base and Zn-base anode materials during production of sacrificial anodes. The results are *not* applicable to cathodic protection design because electrochemical data from short-term testing will tend to be non-conservative with respect to long-term performance in the field.

A.1.2 The Operator may require that testing shall be carried out in a laboratory independent of the anode manufacturer.

A.2 SAMPLING AND PREPARATION OF TEST SPECIMENS

A.2.1 Unless otherwise specified by the Operator, quality control of electrochemical properties shall be carried out for each 15 tonnes of anodes ('batch') produced.

A.2.2 Specimens for testing may be either cut from a produced anode of each batch or cast separately.

A.2.3 One cylindrical specimen with a diameter of 10 ± 1 mm and a length of 50 ± 5 mm is to be machined from each sample material. Adequate marking is to be maintained throughout preparation and testing.

A.2.4 A hole (about 2 mm diameter) is to be drilled and tapped on one of the ends for connection of a support rod in titanium. The support rod shall have an electrically insulating coating on all areas that are exposed to the test solution.

A.2.5 Prior to exposure, the test specimens are to be rinsed in tap water and then in ethanol or equivalent. After drying, the specimens are to be weighed to the nearest 0.1 mg.

A.3 EQUIPMENT AND EXPERIMENTAL PROCEDURE

A.3.1 After mounting the support rod, the specimen ends are to be coated by a suitable sealing compound such as neoprene glue or silicone rubber.

A.3.2 The testing solution may be natural seawater, artificial seawater according to ASTM D1141-52 (constituents other than Cl^- , SO_4^{2-} , $\text{HCO}_3^-/\text{CO}_3^{2-}$, Na^+ , Ca^{2+} and Mg^{2+} may be deleted) or 3% (weight) sodium chloride solution.

A.3.3 A minimum of 10 liters per specimen shall be used. The solution shall be continuously purged with air and shall not be exchanged during the test period. The temperature shall be maintained at $20 \pm 3^\circ\text{C}$.

A.3.4 Each specimen shall be suspended in the centre of an uncoated cylindrical steel screen or pipe segment, the surface area of which shall be minimum 20 times the anode specimen area, i.e. minimum 400 cm^2 approximately. The general arrangement is shown in Fig. A.1.

A.3.5 Galvanostatic control shall be effected by means of a D.C. constant current source. The specimen and the steel cathode are to be coupled to the positive ('Hi') and negative ('Lo') terminals respectively. Multiple testing cells may be connected in series to one D.C. source.

A.3.6 A current integrator, e.g. a copper coulometer or some electronic device capable of determining the total discharged current to an accuracy of $\pm 2\%$, shall be inserted into the circuit containing one or more test cells arranged in series.

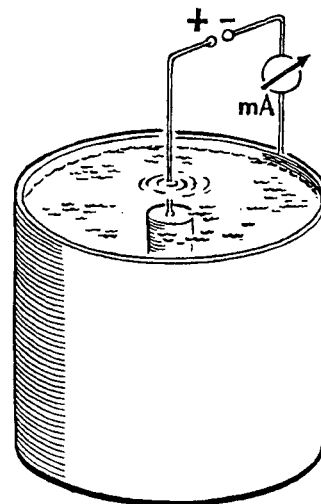


Fig. A1. General arrangement for quality control testing of sacrificial anode materials.

A.3.7 The current through the cell(s) shall be adjusted to provide anodic current densities, based on the initial exposed specimen surface area, in accordance with the following scheme:

- Day 1: 1.5 mA/cm^2
- Day 2: 0.4 mA/cm^2
- Day 3: 4.0 mA/cm^2
- Day 4: 1.5 mA/cm^2

The current density shall be controlled to an accuracy of $\pm 0.1\text{ mA/cm}^2$ and shall be maintained for a period of 24 ± 1 hours. The total testing time shall be 96 ± 4 hours.

A.3.8 At the end of each period (A.3.7), the anode potential shall be measured at 3 positions per specimen. A standard reference electrode with an electrolyte bridge (e.g. a Luggin capillary) is to be applied. The tip of the bridge is to be positioned within 1 mm from the specimen surface. The specimen shall not be cleaned of corrosion products prior to measurements.

Where a reference electrode other than Ag/AgCl/seawater is used, the reference electrode type shall be stated with the recorded reading, together with the reading converted to Ag/AgCl/seawater.

A.3.9 After completion of the full test period, the specimens shall be cleaned of corrosion products. Al-base specimens are to be cleaned for 10 min. at 80°C in a solution containing 20g chromium trioxide and 30 ml conc. phosphoric acid per liter water. Zn-base specimens shall be immersed for 2 hours in a saturated ammonium chloride solution at ambient temperature. Specimens are subsequently to be rinsed in tap water, thereafter in ethanol or equivalent, dried and weighed to the nearest 0.1 mg.

A.3.10 The electrochemical efficiency (ϵ Ah/kg) is to be calculated from

$$\epsilon = \frac{C \cdot 1000}{W}$$

where C is the total current charge in ampere-hours and W is the specimen weight-loss in grams.

A.4 ACCEPTANCE CRITERIA AND RE-TESTING

A.4.1 The measured potentials and calculated efficiency shall meet the requirements for *quality control* stipulated in the anode manufacturing specification.

Typical values are 2300 to 2700 Ah/kg and 750 to 780 Ah/kg for Al-base and Zn-base anode materials respectively.

A.4.2 In case of failure to meet the specified requirements, re-tests may be carried out on 3 specimens from each heat of the batch (see A.2.1). Failure of any specimen to meet the specified requirements shall lead to rejection of all anodes from the heat(s) involved.

A.5 DOCUMENTATION

A.5.1 The test report shall contain all relevant data for the test material (heat number, chemical composition and casting date). Specimen preparation, test equipment and procedures shall be outlined.

A.5.2 All recorded potentials shall be presented in a tabular form. Anode weight loss and total current charge shall be given in addition to the calculated electrochemical efficiency.

APPENDIX B

GUIDELINES FOR LONG-TERM TESTING OF SACRIFICIAL ANODE MATERIALS WITH THE OBJECTIVE OF PERFORMANCE DOCUMENTATION

B.1 GENERAL

B.1.1 The performance of sacrificial anode materials is best documented by field experience, e.g. by utilization of monitored anodes, or by dedicated field testing of full size anodes. Apart from testing in ambient seawater, long term testing may be performed to verify anode performance in special environments, e.g. hot seawater or sediments.

This Appendix gives a simplified procedure for documentation of anode performance in ambient temperature sea water using small-scale testing in a laboratory with access to natural seawater.

B.1.2 Testing for documentation of anode performance is preferably to be carried out in a laboratory independent of the anode manufacturer.

B.2 SAMPLING AND PREPARATION OF TEST SPECIMENS

B.2.1 Material for testing is to be sampled either from manufactured anodes or from special castings using the same raw materials and smelting practice as for normal production.

Duplicate specimens are to be prepared from materials representing a minimum of 5 heats. The chemical composition of the material test samples shall be representative for normal production, i.e. the alloying and impurity element contents shall reflect manufacturer's internal compositional limits for their product.

B.2.2 Cylindrical specimens with a diameter of minimum 20 mm and a length of minimum 80 mm are to be machined from the sampled material.

B.2.3 A hole (about 2 mm diameter) is to be drilled and tapped on one of the ends for connection of a support rod in titanium. The support rod shall have an electrically insulating coating on all areas that are exposed to the test solution.

B.2.4 Prior to exposure, the test specimens are to be rinsed in tap water and then in ethanol or equivalent. After drying, the specimens are to be weighed to the nearest 0.1 mg.

B.2.5 After mounting the support rod, the specimen ends are to be coated by a suitable sealing compound such as neoprene glue or silicone rubber.

B.3 TESTING EQUIPMENT AND EXPERIMENTAL PROCEDURE

B.3.1 The testing environment shall be clean natural seawater with a salinity ≥ 30 ‰. Temperature may be controlled or may be allowed to vary within the range $+7^{\circ}\text{C}$ to $+20^{\circ}\text{C}$. Temperature and salinity are to be recorded at least once weekly.

B.3.2 The test cells shall be configured as outlined in Appendix A, but there shall be a continuous exchange of seawater in the cell (minimum 1 l/min). The seawater is to be continuously purged with air.

B.3.3 Testing may be performed with either galvanostatic control or as a "free-running test". The latter implies that the anode/cathode assembly is allowed to polarize spontaneously during the test. The driving voltage, and hence the anodic current density, will thus vary during the test.

The testing period shall be minimum 12 months.

B.3.4 With **galvanostatic control**, the anodic current density shall be 1 mA/cm² based on the initial exposed specimen surface area. More complete information documentation will be achieved by extended testing in the range 0.4 to 4 mA/cm².

The integrated current may be determined as in Appendix A, but manual readings minimum once per week will suffice.

Recordings of anode potentials (see Appendix A), shall be performed minimum once per day and 5 days per week during the first 2 weeks of testing, thereafter minimum once per week.

B.3.5 In the case of the **free-running test** procedure, the anode and cathode are to be connected over a 10 ohm precision resistance in order to limit the anodic current density. Higher and lower circuit resistances (e.g. 2-20 ohm) may be applied when more extensive documentation is desired. The cathode surface area shall be minimum 20 times the exposed anode surface area.

B.3.6 The anodic current density is to be calculated from measurements of the voltage drop across the 10 ohm circuit resistance using a high impedance ($\geq 10^7$ ohm) voltmeter.

Measurements of current and anode potential are to be made at intervals as specified in B.3.4. For free-running tests the potential of the steel cathode shall also be recorded at the same time.

B.3.7 For both galvanostatic and free-running tests, the electrochemical efficiency shall be calculated as specified in Appendix A.

B.4 DOCUMENTATION

B.4.1 The test report shall contain all relevant data from smelting and casting of the test material, including the contents of all alloying and impurity elements affecting anode performance. Nominal composition and guaranteed compositional limits shall be given as a reference.

B.4.2 Specimen preparation, testing equipment and procedures are to be detailed. Environmental data including seawater temperature and salinity shall be reported.

B.4.3 Anode potentials (versus Ag/AgCl/seawater) shall be presented graphically as a function of time. For free running tests the calculated anodic current density and the cathode potential shall also be given in graphical form.

Photographs of anode specimens prior to, and after cleaning shall be included. Electrochemical efficiency shall be given in A·h/kg.

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